

Introduction:

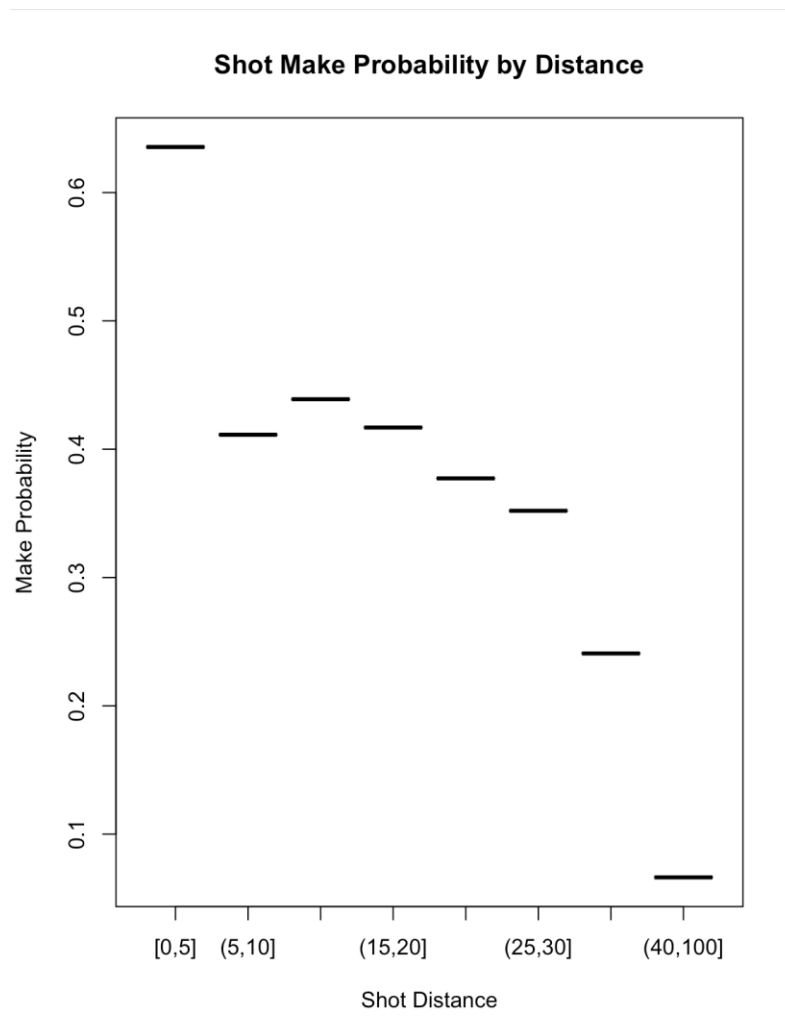
The objective of this project was to predict the probability that an NBA shot would be made using player tracking and shot-level features. The training dataset contains 425,719 regular-season shot attempts, while the testing dataset contains 213,977 shots. The final model produces a probability between 0 and 1 for each shot in the testing dataset. I wanted to focus on producing well-calibrated shot probabilities rather than only predicting whether a certain shot would be a make or miss.

The dataset contains several types of information describing each shot, including shot location, distance, shot type, defensive pressure, shooter movement, game context, and shot-clock information. I thought it was important for me to examine the structure and distributions of the variables to get familiar with the data. I compared the training and testing datasets for categorical variables and did checks that found no duplicate rows in either dataset, and the missing-value patterns in the training and testing data were similar.

For my model evaluations, I split the training data into an 80% training set and a 20% validation set using a stratified split. This helped me evaluate the models on observations that were not used during model fitting.

Exploratory Analysis:

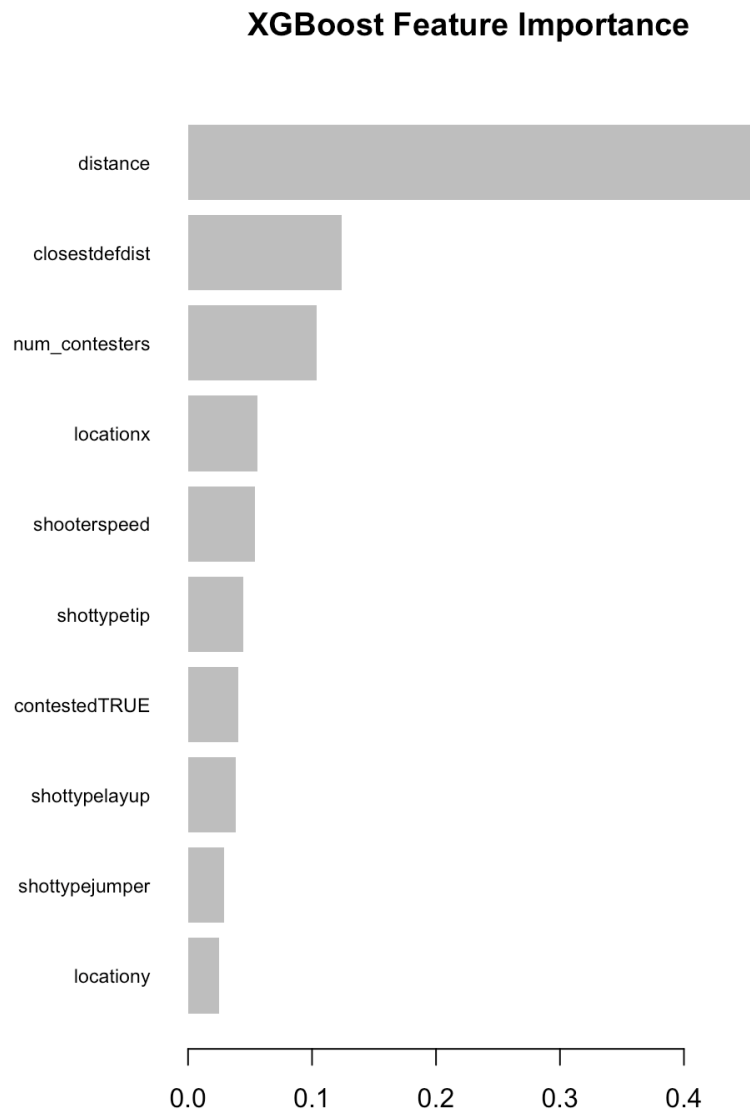
The first relationship I looked at was the relationship between shot distance and shot-making probability. I grouped shots into intervals based on distance and calculated the make rate within each interval. What was shown was that shot distance was strongly related to shot-making probability.



The first model out of three that I created uses shot distance to estimate the probability of a made shot. The fitted model showed a negative relationship between distance and the probability of making a shot. The estimated odds ratio for distance was about 0.957, which means that a one foot increase in distance caused about a 4.3% decrease in the odds of a make in this model.

For my second model, I modified the logistic regression model to include more characteristics of the shot such as shot distance, x and y shot location, shooter speed, number of dribbles before the shot, closest defender distance, number of contesters, whether the shot was contested, whether the shot was a three-point attempt, and shot type. This model was an experiment to see if adding more context to the shot improved predictive performance.

For my third model, I wanted to allow more nonlinear relationships between the predictors and shot outcomes. For this model, I used XGBoost, which was able to highlight nonlinear relationships between variables without requiring those relationships to be manually specified.



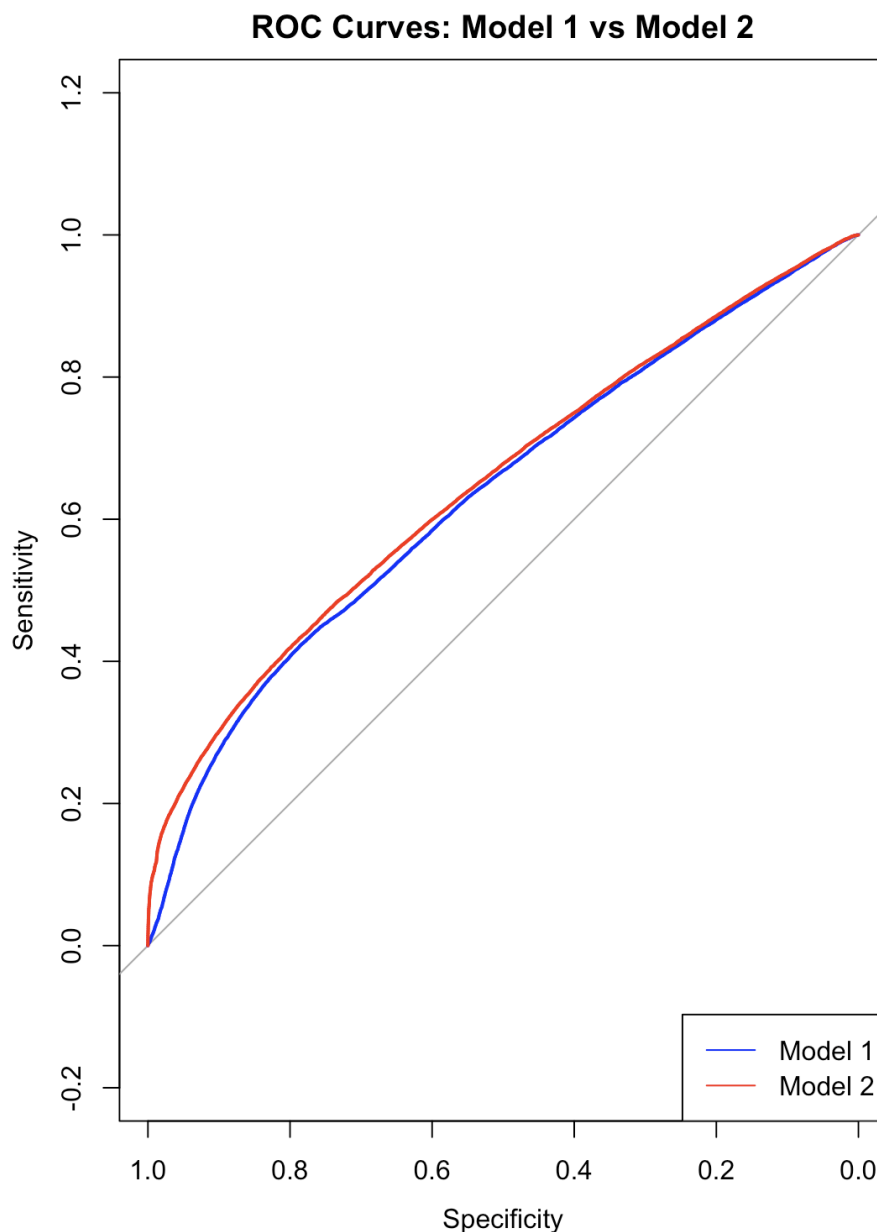
Model Evaluation:

When evaluating my models, I used AUC, log loss, accuracy, sensitivity, specificity, and precision. The results show an improvement in AUC and log loss as more features were added.

For Model 1, it has an AUC of 0.6333 and a log loss of 0.6657. Adding variables increased the AUC to 0.6484 and decreased log loss to 0.6421. Model 3 had the strongest validation results, with an AUC of 0.6653 and a log loss of 0.6289.

Model 3 also had the highest accuracy and precision at a 0.5 classification threshold and the highest specificity. However, its sensitivity was lower than the other two logistic regression models. This is why I focused on probability-based evaluations.

The ROC curves show a threshold-independent comparison of the models. Model 3 had the highest AUC at 0.6653, compared with 0.6484 for Model 2 and 0.6333 for Model 1. The improvement shows that the additional features and nonlinear modeling approach allowed Model 3 to distinguish between made and missed shots more effectively.



I believe Model 3 is the best model to use for the final testing predictions because it produced the lowest validation log loss and highest AUC out of the three models. The XGBoost model was then used to generate a predicted probability for every shot in the testing dataset. The testing dataset contains 213,977 observations, and my prediction also contains 213,977 values.

Final Prediction:

The predictions were then placed into the required "make_prob" column of the submission template. The predicted probabilities ranged from about 0.015 to 0.993, with a mean of

approximately 0.456 based on the Model 3 testing predictions. The final submission contained 213,977 rows, matching the number of testing observations.

Improvements:

While working on this project, I noticed that the dataset contains a “gamestate” variable which includes context (halfcourt, transition, and second chance, as well as shot clock). These variables could possibly improve predictions because the same shot location may have different characteristics depending on the possession. Also, the dataset contains a “shooter_id” variable, which could be used to catch differences in individual shooting ability. A potential tweak to this would be to incorporate historical shooter performance, while making sure that only information available before the shot is used.

Conclusion:

I developed three models to predict NBA shot making probabilities. The first model established a baseline using shot distance. The second model incorporated spatial, shooting, and defensive variables, while the third model used XGBoost to capture more flexible relationships among those predictors.

Performance improved as the modeling approach became more detailed. Model 3 achieved the lowest validation log loss (0.6289) and highest AUC (0.6653) among the three models evaluated. Because the project evaluates the submitted probabilities using log loss, I used Model 3 to generate the final predictions for the testing dataset. Overall, the modeling process progressed from a simple interpretable baseline toward a more complex machine-learning model.